Agriculture Sciences 9 — Solved Exercises (PCTB)

Class 9 — Solved Exercises

Punjab Curriculum and Textbook Board (PCTB)

10 chapters — 183 exercises

Chapter 1: Agriculture in Pakistan

Note on this book

Note: how the Inquisitive Questions are answered

The book sets Inquisitive Questions as open-ended thinking tasks and supplies no answer key for them. In this book they are answered as clearly labelled model answers, worked from the chapter's own facts, in the same way as the open-ended tasks in earlier books. Definite questions (MCQs, Short and Long Questions) are answered directly.

Multiple Choice Questions (MCQs)

1. What is the primary purpose of agriculture?

(a) Industrial development

(b) Food production ✓

(c) Urbanization

(d) Tourism

Correct answer: Food production

Multiple Choice Questions (MCQs)

2. Which of the following is NOT a branch of agriculture?

(a) Agronomy

(b) Horticulture

(c) Aquaculture

(d) Astronomy ✓

Correct answer: Astronomy

Multiple Choice Questions (MCQs)

3. Which crop is a staple food in Pakistan?

(a) Wheat ✓

(b) Coffee

(c) Tea

(d) Cotton

Correct answer: Wheat

Multiple Choice Questions (MCQs)

4. What is the main source of water for irrigated agriculture in Pakistan?

(a) Rainfall

(b) Indus River System ✓

(c) Groundwater only

(d) Desalination plants

Correct answer: Indus River System

Multiple Choice Questions (MCQs)

5. Which of the following is a rain-fed crop in Pakistan?

(a) Rice

(b) Sugarcane

(c) Barley ✓

(d) Cotton

Correct answer: Barley

Multiple Choice Questions (MCQs)

6. What is the primary role of agroforestry?

(a) Increasing urbanization

(b) Integrating trees with crops and livestock ✓

(c) Reducing industrial output

(d) Promoting tourism

Correct answer: Integrating trees with crops and livestock

Multiple Choice Questions (MCQs)

7. Which of the following is a major constraint in Pakistani agriculture?

(a) Excessive rainfall

(b) Water scarcity ✓

(c) Over-irrigation

(d) Lack of labor

Correct answer: Water scarcity

Multiple Choice Questions (MCQs)

8. What is the main purpose of crop rotation?

(a) Increasing soil erosion

(b) Maintaining soil fertility ✓

(c) Reducing crop diversity

(d) Increasing water usage

Correct answer: Maintaining soil fertility

Multiple Choice Questions (MCQs)

9. Which of the following is a high-value crop in Pakistan?

(a) Wheat

(b) Mango ✓

(c) Barley

(d) Millet

Correct answer: Mango

Multiple Choice Questions (MCQs)

10. What is the primary benefit of drip irrigation?

(a) Increased water wastage

(b) Reduced water usage ✓

(c) Higher soil erosion

(d) Increased labor costs

Correct answer: Reduced water usage

Short Questions

1. What is the difference between irrigated and rain-fed agriculture?

Irrigated agriculture supplies water to crops artificially, through canals, tube wells or other irrigation systems, so cropping does not depend on rainfall. Rain-fed (barani) agriculture depends entirely on rainfall for crop water, so it is possible only where rainfall is sufficient and reliable, and yields vary with the rains. Irrigated farming gives higher and more reliable yields; rain-fed farming is cheaper but riskier.

Short Questions

2. What are the main environmental benefits of agroforestry?

Agroforestry protects the environment in several ways: tree roots bind the soil and reduce erosion; trees store carbon and help fight climate change; trees and mixed planting support biodiversity by providing habitats for birds, bees and beneficial insects; and tree cover improves water retention in the soil and reduces runoff. It also reduces the need for chemical inputs, because leaf litter adds organic matter to the soil.

Short Questions

3. What is the role of poultry farming in Pakistan's economy?

Poultry farming supplies cheap protein in the form of meat and eggs for a growing population, provides employment to millions of people in farming, feed production, processing and sale, and supports related industries such as feed mills and hatcheries. It is one of the fastest-growing and most organised parts of

Pakistan's agriculture, and it strengthens food security, especially for low-income families.

Long Questions

1. Discuss the importance of agriculture in Pakistan's economy, focusing on its contribution to GDP, employment, and food security.

Agriculture is the backbone of Pakistan's economy. It contributes a large share of GDP directly through crops and livestock, and a still larger share indirectly through agro-based industries such as textiles, sugar and food processing. It employs a very large part of the labour force, especially in rural areas where farming is often the only livelihood. For food security, agriculture produces the country's staple foods, wheat, rice, pulses, milk and meat, so the population's daily food depends on it. It also earns foreign exchange through exports of cotton, rice, fruits and other products, and supplies raw materials to industry. Without a productive agriculture, Pakistan would face food shortages, unemployment and a weaker economy.

Long Questions

2. What are the major constraints faced by farmers in Pakistan? How can these constraints be addressed to improve agricultural productivity?

The major constraints are water scarcity and inefficient irrigation, small and fragmented landholdings, poor-quality seed and low use of modern technology, soil degradation and salinity, expensive inputs such as fertiliser and diesel, weak access to credit and markets, and climate change with erratic rainfall and heat. They can be addressed by lining watercourses and using drip and sprinkler irrigation, supplying certified seed and training farmers through extension services, reclaiming saline soils and promoting crop rotation, giving affordable credit and fair market access, and developing climate-resilient crop varieties.

Government support, research and farmer education together can raise productivity.

Long Questions

3. Compare and contrast irrigated and rain-fed agriculture in Pakistan, highlighting their advantages and disadvantages.

Irrigated agriculture uses canal or tube-well water, so crops can be grown in dry areas and in both Rabi and Kharif seasons with high, reliable yields; its disadvantages are high cost, waterlogging and salinity where drainage is poor, and dependence on the canal system. Rain-fed agriculture depends only on rainfall, so it costs little and suits hilly and barani areas; but yields are low and uncertain, only drought-tolerant crops such as barley, pulses and oilseeds do well, and a failed rain can mean a failed crop. Irrigated farming feeds the bulk of the country, while rain-fed farming supports millions of small farmers in areas canals cannot reach.

Inquisitive Questions

1. How can modern technology, such as precision agriculture and biotechnology, revolutionize farming practices in Pakistan?

Model answer: precision agriculture uses sensors, drones, satellite images and soil testing to apply exactly the right amount of water, fertiliser and pesticide where each part of a field needs it, cutting waste and raising yields. Biotechnology provides improved varieties that resist pests and diseases, tolerate drought and salinity, and give higher yields. Together with mobile phone advisories and weather forecasts, these tools let even small farmers plan sowing, irrigation and harvesting better, reduce costs, and protect the soil and water for the future.

Inquisitive Questions

2. What role can the government play in promoting sustainable agricultural practices, such as agroforestry and organic farming, in Pakistan?

Model answer: the government can promote sustainable practices by giving subsidies and training for agroforestry and organic farming, supplying quality saplings and organic inputs, and paying farmers for planting trees on their land. It can set up research and extension services that teach composting, crop rotation and integrated pest management, enforce laws against deforestation and excessive pesticide use, and help farmers market organic and agroforestry products at better prices. Public investment in water conservation and soil reclamation would support the same goal.

Chapter 2: Crop Production and Farming System

Multiple Choice Questions (MCQs)

1. What is the primary goal of crop rotation?

(a) Increase soil erosion

(b) Maintain soil fertility and reduce pests ✓

(c) Grow the same crop every year

(d) Reduce biodiversity

Correct answer: Maintain soil fertility and reduce pests

Multiple Choice Questions (MCQs)

2. Which crop is known for adding nitrogen to the soil?

(a) Wheat

(b) Rice

(c) Legumes (e.g., beans, peas) ✓

(d) Corn

Correct answer: Legumes (e.g., beans, peas)

Multiple Choice Questions (MCQs)

3. What is a common cropping pattern in the Indus Plains of Pakistan?

(a) Wheat-Rice rotation ✓

(b) Maize-Potato rotation

(c) Barley-Fruit rotation

(d) Cotton-Sugarcane rotation

Correct answer: Wheat-Rice rotation

Multiple Choice Questions (MCQs)

4. Which factor does NOT affect crop production?

(a) Climate

(b) Market demand

(c) Soil type

(d) Farmer's favorite color ✓

Correct answer: Farmer's favorite color

Multiple Choice Questions (MCQs)

5. What is the main benefit of intercropping?

(a) Increases soil erosion

(b) Reduces biodiversity

(c) Optimizes resource use and reduces pests ✓

(d) Requires less labor

Correct answer: Optimizes resource use and reduces pests

Multiple Choice Questions (MCQs)

6. Which region in Pakistan is known for fruit orchards like apples and apricots?

(a) Indus Plains

(b) Coastal Areas

(c) Mountainous Regions (Khyber Pakhtunkhwa, Gilgit-Baltistan) ✓

(d) Arid Deserts

Correct answer: Mountainous Regions (Khyber Pakhtunkhwa, Gilgit-Baltistan)

Short Questions

1. What is crop rotation, and why is it important?

Crop rotation is the practice of growing different crops on the same field in a planned sequence across seasons, instead of repeating the same crop. It is important because different crops use different nutrients, so rotation maintains soil fertility; it breaks the life cycles of pests and diseases that attack one crop; legumes in the rotation add nitrogen to the soil; and it improves soil structure

and reduces erosion.

Short Questions

2. Name two crops that are commonly grown in a wheat-rice rotation.

The two crops of the wheat-rice rotation are wheat itself, grown as the Rabi (winter) crop, and rice, grown as the Kharif (summer) crop. This is the dominant rotation of the irrigated Indus Plains of Punjab and Sindh.

Short Questions

3. How do legumes benefit the soil in a crop rotation system?

Legumes such as beans, peas, lentils and clover carry nitrogen-fixing bacteria in nodules on their roots. These bacteria take nitrogen from the air and convert it into forms plants can use, enriching the soil with nitrogen. The following crop then needs less nitrogen fertiliser, and the soil's organic matter and structure also improve.

Short Questions

4. What is the main challenge of farming in arid regions of Pakistan?

The main challenge is shortage of water: rainfall is low and unreliable, and there are few canals or tube wells. Soils are often poor and saline, and hot, dry winds increase evaporation. Farmers must therefore grow drought-tolerant crops, conserve every drop of water, and protect the soil from wind erosion.

Short Questions

5. What is intercropping, and how does it help farmers?

Intercropping is growing two or more crops together on the same field at the same time, for example maize intercropped with beans or vegetables. It helps farmers because the crops use light, water and nutrients differently, so the land

produces more in total; the mixture suppresses weeds and reduces pest and disease attack; and if one crop fails, the other still gives an income, spreading the farmer's risk.

Short Questions

6. Why is terracing used in mountainous regions for farming?

Terracing cuts steep slopes into a series of flat steps. The flat steps slow down running water, so rain does not wash the soil away, and water soaks into the soil instead of running off. Terracing therefore prevents soil erosion on hillsides and creates level land where crops can be grown and irrigated, as in the mountainous regions of Khyber Pakhtunkhwa and Gilgit-Baltistan.

Short Questions

7. What is the limitation of Mono-Cropping?

Mono-cropping, growing the same crop on the same field year after year, depletes the same nutrients from the soil, encourages the build-up of pests and diseases that attack that crop, and degrades soil structure. Yields fall over time, the farmer must use more fertiliser and pesticide, and a single crop failure or price fall can ruin his whole income.

Long Questions

1. Explain the economic significance of major crops grown in Pakistan.

Pakistan's major crops drive the whole rural economy. Wheat, the staple food, ensures food security and employs millions in growing, milling and trade. Cotton, called white gold, feeds the textile industry, Pakistan's largest export earner. Rice, especially Basmati, is a major export crop earning foreign exchange. Sugarcane supports the sugar industry, and maize, pulses and oilseeds feed people and livestock. Together these crops provide income to farming families,

raw material to industry, and exports that support the national economy.

Long Questions

2. How do climate and water availability play a role in cropping pattern?

Climate decides which crops can grow where: temperature, rainfall and season length fix the Rabi crops of winter (wheat, barley, gram) and the Kharif crops of summer (rice, cotton, sugarcane, maize). Water availability decides the pattern even more sharply: irrigated plains grow water-demanding rice and sugarcane, rain-fed areas grow barley, pulses and oilseeds, and arid regions can grow only drought-tolerant crops. Where water is abundant, farmers choose high-value, high-water crops; where it is scarce, they choose hardy, low-water crops and fallowing.

Inquisitive Questions

1. If you were a farmer in a water-scarce region, which cropping pattern would you choose and why?

Model answer: in a water-scarce region I would choose a drought-tolerant pattern: rain-fed barley, pulses such as gram and lentils, and oilseeds, rotated with a legume to restore soil nitrogen, and intercropped to use the land fully. I would avoid water-hungry rice and sugarcane, use mulching and rainwater harvesting to keep moisture in the soil, and keep part of the land fallow in dry years. This pattern gives a reliable, if modest, harvest with the little water available, instead of risking total failure on a thirsty crop.

Inquisitive Questions

2. How can modern technology improve the efficiency of crop rotation and intercropping systems?

Model answer: modern technology improves these systems through soil testing

and satellite mapping, which show exactly which nutrients each part of a field lacks, so the rotation can include the right legume or cover crop. Mobile advisories and weather forecasts help farmers time sowing and harvesting in the rotation. Precision planters sow intercrops at the right spacing, drones monitor pest attack in mixed crops, and improved varieties bred for intercropping give higher combined yields. Record-keeping apps also let farmers plan rotations over several years instead of guessing.

Chapter 3: Production Technologies of Major Rabi and Kharif Crops

Multiple Choice Questions (MCQs)

1. What is the ideal temperature range for wheat growth?

(a) 5 degrees C - 10 degrees C

(b) 10 degrees C - 25 degrees C ✓

(c) 30 degrees C - 40 degrees C

(d) 40 degrees C - 50 degrees C

Correct answer: 10 degrees C - 25 degrees C

Multiple Choice Questions (MCQs)

2. Which crop is known as "white gold" in Pakistan?

(a) Wheat

(b) Rice

(c) Cotton ✓

(d) Sugarcane

Correct answer: Cotton

Multiple Choice Questions (MCQs)

3. What is the best soil type for potato cultivation?

(a) Sandy loam ✓

(b) Clay

(c) Rocky soil

(d) Saline soil

Correct answer: Sandy loam

Multiple Choice Questions (MCQs)

4. Which of the following is a Rabi crop?

(a) Rice

(b) Maize

(c) Wheat ✓

(d) Cotton

Correct answer: Wheat

Multiple Choice Questions (MCQs)

5. What is the critical irrigation stage for sugarcane?

(a) Germination

(b) Tillering

(c) Grand growth

(d) All of the above ✓

Correct answer: All of the above

Note on Question 5

Note: why is 'all of the above' the answer for sugarcane irrigation?

Sugarcane needs adequate water at every stage: germination to establish the crop, tillering to produce enough stalks, and grand growth when the cane grows fastest and uses the most water. No single stage can be singled out, because water stress at any of them cuts the yield; the grand-growth stage is, however, the most water-sensitive of the three.

Multiple Choice Questions (MCQs)

6. Which fertilizer is essential for root development in crops?

(a) Nitrogen

(b) Phosphorus ✓

(c) Potassium

(d) Zinc

Correct answer: Phosphorus

Multiple Choice Questions (MCQs)

7. Which crop is used for both food and brewing purposes?

(a) Barley ✓

(b) Wheat

(c) Rice

(d) Maize

Correct answer: Barley

Multiple Choice Questions (MCQs)

8. What is the main pest of cotton?

(a) Aphids

(b) Bollworms ✓

(c) Termites

(d) Armyworms

Correct answer: Bollworms

Multiple Choice Questions (MCQs)

9. Which crop is grown during the Kharif season?

(a) Wheat

(b) Barley

(c) Rice ✓

(d) Mustard

Correct answer: Rice

Multiple Choice Questions (MCQs)

10. What is the ideal pH range for rice cultivation?

(a) 4.0 - 5.0

(b) 5.5 - 7.5

(c) 8.0 - 9.0

(d) 9.0 - 10.0

Correct answer: 5.5 - 7.5

Short Questions

1. What are Rabi crops? Give two examples.

Rabi crops are sown in winter (October to December) and harvested in spring (April to May), growing on the cool season's moisture and irrigation. Two examples are wheat, Pakistan's main staple, and barley; other Rabi crops are gram, mustard and lentils.

Short Questions

2. Why is cotton called "white gold"?

Cotton is called white gold because its white lint is extremely valuable: it is the raw material of Pakistan's textile industry, the country's largest industry and export earner, and it provides income to millions of farming families and textile workers. Like gold, it is a major source of national wealth.

Short Questions

3. What is the importance of seed treatment in crop production?

Seed treatment means coating seed with fungicide, insecticide or biofertiliser before sowing. It protects the seed and young seedling from soil-borne diseases and insects, improves germination and early growth, and reduces the need for later pesticide sprays. Healthy seedlings from treated seed give a uniform crop stand and higher yields.

Short Questions

4. Name two diseases of wheat and their control measures.

Two common wheat diseases are rust (brown, yellow or black rust) and smut. Rust is controlled by growing resistant varieties and, when needed, spraying fungicide at the first sign of disease. Smut is controlled by seed treatment with fungicide before sowing and by removing and destroying infected plants, so the disease does not spread to the next crop.

Short Questions

5. What is the role of phosphorus in plant growth?

Phosphorus is essential for root development, especially in young plants, and for energy transfer inside the plant, flowering, seed formation and early maturity. Plants short of phosphorus have weak roots, stunted growth and poor grain formation, which is why phosphatic fertilisers such as superphosphate are applied at sowing time.

Short Questions

6. Why is irrigation critical during the flowering stage of sugarcane?

The question as printed asks about the flowering stage, but sugarcane grown for sugar is normally harvested before it flowers, and its truly critical stages are germination, tillering and grand growth, as in MCQ 5. In general, water stress at flowering in any crop causes flowers to drop and grains to fail, so irrigation at flowering protects the yield. For sugarcane itself, the grower must never let the crop suffer water shortage during its grand-growth stage, when it is building most of its stalk weight.

Long Questions

1. Explain the production technology of wheat, including land preparation,

sowing, and fertilizer management.

Land preparation: after the Kharif crop, the field is ploughed two or three times and levelled to make a fine, firm seedbed, and farmyard manure is mixed in. Sowing: certified seed of a recommended variety is sown in November at the right seed rate, usually with a drill so the seed is placed at uniform depth and spacing, and seed is treated with fungicide first. Fertiliser management: nitrogen, phosphorus and potash are applied on the basis of soil testing; phosphorus and potash go in at sowing, while nitrogen is split, part at sowing and the rest with the first and second irrigations at tillering and stem extension. Weeds are controlled, and four to five irrigations are given at crown-root initiation, tillering, jointing, flowering and grain filling, before harvest in April or May.

Long Questions

2. Discuss the importance of Kharif crops in Pakistan's economy and their role in food security.

Kharif crops, grown in the summer monsoon season, include rice, cotton, sugarcane and maize. Rice, especially Basmati, is a leading export earner and a staple food in many areas. Cotton feeds the textile industry and millions of jobs. Sugarcane supplies the sugar industry, and maize feeds people and livestock and supports the poultry feed industry. Together Kharif crops provide summer employment, raw materials for industry, export income and a large part of the country's food and fodder, so a good Kharif harvest is essential for food security and for the whole economy.

Long Questions

3. Describe the steps involved in the production of rice, from land preparation to post-harvest management.

Land preparation: the field is ploughed and then puddled, that is ploughed again under standing water, to make a soft mud that holds water and controls weeds, and it is levelled. Nursery: seed is sown in a small nursery bed and seedlings are raised for about a month. Transplanting: seedlings are transplanted into the flooded field in rows. Growth care: the field is kept flooded, nitrogen fertiliser is split into doses, and weeds, stem borers and blast disease are controlled. Harvest: the crop is cut when the grains are mature and golden, threshed, and the paddy is dried to safe moisture before storage or milling, so it does not spoil or lose quality.

Inquisitive Questions

1. How can modern technology, like drones and sensors, improve crop yields in Pakistan?

Model answer: drones fitted with cameras can survey a whole field in minutes, spotting pest attack, disease patches and water stress before the human eye can, so the farmer treats only the affected spots. Soil-moisture sensors tell exactly when and how much to irrigate, saving water and avoiding stress. Satellite images and mobile advisories guide sowing dates and fertiliser doses, and GPS-guided machinery sows and sprays evenly. Together these tools cut input costs, reduce waste and raise yields, even on small farms, if the services are shared through cooperatives.

Inquisitive Questions

2. What are the environmental impacts of excessive fertilizer use in agriculture, and how can they be minimized?

Model answer: excessive fertiliser pollutes the environment: nitrogen leaches into groundwater and runs off into rivers, causing algal blooms that kill fish; it also escapes as gases that harm the climate; and salts build up in the soil,

damaging its structure and microbes. These impacts can be minimised by soil testing and applying only the dose the crop needs, splitting nitrogen into small doses, using farmyard manure and compost alongside chemical fertiliser, planting legumes to supply nitrogen naturally, and keeping buffer strips of grass or trees along waterways to catch runoff.

Chapter 4: Crop Rotation and Cropping Pattern

Note on this chapter

Note: this chapter's exercise repeats much of Chapter 2's exercise

The book prints almost the same exercise for this chapter as for Chapter 2: MCQs 1-5 and Short Questions 1-3 are identical in wording, and the remaining items cover the same topics. They are solved here in full under this chapter, as printed, rather than cross-referred, so that this chapter's exercise is complete on its own. The answers are naturally the same where the questions are the same.

Multiple Choice Questions (MCQs)

1. What is the primary goal of crop rotation?

(a) Increase soil erosion

(b) Maintain soil fertility and reduce pests ✓

(c) Grow the same crop every year

(d) Reduce biodiversity

Correct answer: Maintain soil fertility and reduce pests

Multiple Choice Questions (MCQs)

2. Which crop is known for adding nitrogen to the soil?

(a) Wheat

(b) Rice

(c) Legumes (e.g., beans, peas) ✓

(d) Corn

Correct answer: Legumes (e.g., beans, peas)

Multiple Choice Questions (MCQs)

3. What is a common cropping pattern in the Indus Plains of Pakistan?

(a) Wheat-Rice rotation ✓

(b) Maize-Potato rotation

(c) Barley-Fruit rotation

(d) Cotton-Sugarcane rotation

Correct answer: Wheat-Rice rotation

Multiple Choice Questions (MCQs)

4. Which factor does not affect cropping patterns?

(a) Climate

(b) Market demand

(c) Soil type

(d) Farmer's favorite color ✓

Correct answer: Farmer's favorite color

Multiple Choice Questions (MCQs)

5. What is the main benefit of intercropping?

(a) Increases soil erosion

(b) Reduces biodiversity

(c) Requires less labor

(d) Optimizes resource use and reduces pests ✓

Correct answer: Optimizes resource use and reduces pests

Multiple Choice Questions (MCQs)

6. Which practice helps prevent soil erosion in hilly areas?

(a) Overgrazing

(b) Deforestation

(c) Terraced farming ✓

(d) Mono cropping

Correct answer: Terraced farming

Short Questions

1. What is crop rotation, and why is it important?

Crop rotation is the practice of growing different crops on the same field in a planned sequence across seasons, instead of repeating the same crop. It is important because different crops use different nutrients, so rotation maintains soil fertility; it breaks the life cycles of pests and diseases that attack one crop; legumes in the rotation add nitrogen to the soil; and it improves soil structure and reduces erosion.

Short Questions

2. Name two crops that are commonly grown in a wheat-rice rotation.

The two crops of the wheat-rice rotation are wheat itself, grown as the Rabi (winter) crop, and rice, grown as the Kharif (summer) crop. This is the dominant rotation of the irrigated Indus Plains of Punjab and Sindh.

Short Questions

3. How do legumes benefit the soil in a crop rotation system?

Legumes such as beans, peas, lentils and clover carry nitrogen-fixing bacteria in nodules on their roots. These bacteria take nitrogen from the air and convert it into forms plants can use, enriching the soil with nitrogen. The following crop then needs less nitrogen fertiliser, and the soil's organic matter and structure also improve.

Short Questions

4. What is the role of market demand in determining cropping patterns?

Market demand strongly shapes what farmers grow: if a crop fetches a good price and sells easily, farmers plant more of it, while a crop with no buyers is dropped even if it grows well. Prices of cotton, rice, vegetables and fruits, access

to markets and roads, and contracts with mills or exporters therefore decide cropping patterns almost as much as climate and soil do.

Short Questions

5. How does crop rotation help in pest and disease management?

Most pests and diseases attack only one crop or crop family and survive in the soil or stubble waiting for that crop to return. Rotation starves them: when a different, non-host crop is grown, the pest cannot feed or multiply and its numbers collapse. Rotating also changes the field environment and lets natural enemies build up, so the next time the host crop returns, pest pressure is far lower and less pesticide is needed.

Long Questions

1. Explain the principles of crop rotation.

The principles of crop rotation are: (1) grow crops of different families in sequence, so the same pests and diseases do not build up; (2) include a legume in the rotation to restore soil nitrogen; (3) alternate deep-rooted and shallow-rooted crops, so nutrients are drawn from different soil depths; (4) alternate crops that deplete the soil with crops that restore it, such as green-manure crops ploughed back into the soil; (5) follow a heavy-feeding crop with a light-feeding one; and (6) plan the sequence over several years, fitting each crop to its proper season, so fertility, structure and pest control are all maintained together.

Long Questions

2. Compare the cropping patterns of the Indus Plains and the Mountainous Regions of Pakistan.

The Indus Plains are flat, irrigated and fertile, so their pattern is intensive:

wheat-rice and cotton-wheat rotations, sugarcane, maize and vegetables, with two or even three crops a year from canal water. The Mountainous Regions of Khyber Pakhtunkhwa and Gilgit-Baltistan are steep, cold and mostly rain-fed, so their pattern is extensive and adapted to slopes: maize and potatoes in Kharif, wheat and barley in Rabi, fruit orchards of apricots, cherries and almonds on terraced slopes, and mixed cropping of maize with beans or vegetables. Plains farming is high-input and market-oriented; mountain farming is low-input, terraced and subsistence-oriented, with fruit as its main cash product.

Long Questions

3. Discuss the factors that influence cropping patterns in Pakistan.

Cropping patterns are influenced by physical factors: climate (temperature and rainfall), soil type and fertility, water availability, and topography, which together fix what can grow where. They are also influenced by economic factors: market demand and prices, cost and availability of inputs, landholding size, and access to credit and transport. Social and technical factors matter too: the farmer's knowledge and traditions, availability of labour, government policies and support prices, and new technology such as improved seed and machinery. In Pakistan, irrigation water is often the deciding factor: where canal water reaches, intensive rotations flourish, and where it does not, hardy rain-fed patterns take over.

Inquisitive Questions

1. If you were a farmer in a region where irrigated water is abundantly available, which cropping pattern would you choose and why?

Model answer: with abundant irrigation water I would choose an intensive, high-value pattern: rice in Kharif followed by wheat in Rabi, the classic Indus Plains rotation, with sugarcane on part of the land as a longer cash crop and vegetables on a small area for quick cash income. Abundant water removes the

main limit on yields, so the best strategy is to grow the crops that give the highest return per acre and to crop the land twice a year, while still rotating with a legume or green-manure crop and keeping drainage good, so that waterlogging and salinity do not ruin the soil.

Chapter 5: Soil and Water Conservation

Multiple Choice Questions (MCQs)

1. What is the primary goal of soil and water conservation?

(a) Increase soil erosion

(b) Prevent soil degradation and manage water resources sustainably ✓

(c) Promote deforestation

(d) Reduce agricultural productivity

Correct answer: Prevent soil degradation and manage water resources sustainably

Multiple Choice Questions (MCQs)

2. Which of the following is a modern technique for soil conservation?

(a) Overgrazing

(b) No-till farming ✓

(c) Deforestation

(d) Excessive use of chemical fertilizers

Correct answer: No-till farming

Multiple Choice Questions (MCQs)

3. Which method is used to collect and store rainwater for irrigation?

(a) Overgrazing

(b) Rainwater harvesting ✓

(c) Deforestation

(d) Excessive plowing

Correct answer: Rainwater harvesting

Multiple Choice Questions (MCQs)

4. What is the role of afforestation in watershed management?

(a) Increase soil erosion

(b) Reduce groundwater recharge

(c) Stabilize soil and improve water retention ✓

(d) Promote desertification

Correct answer: Stabilize soil and improve water retention

Multiple Choice Questions (MCQs)

5. Which of the following is a cause of soil degradation?

(a) Crop rotation

(b) Overgrazing ✓

(c) Contour farming

(d) Mulching

Correct answer: Overgrazing

Multiple Choice Questions (MCQs)

6. Which irrigation method is the most water-efficient?

(a) Flood irrigation

(b) Drip irrigation ✓

(c) Manual irrigation

(d) Furrow irrigation

Correct answer: Drip irrigation

Short Questions

1. What is soil degradation.

Soil degradation is the decline in the quality and productive capacity of soil. It happens through erosion by wind and water, loss of organic matter, salinity and waterlogging, nutrient depletion from continuous cropping, and compaction by heavy machinery. Degraded soil holds less water, supports weaker crops and

gives lower yields.

Short Questions

2. How does contour farming help in soil and water conservation?

In contour farming, crops are planted in rows that follow the natural contours, the level lines, of a slope, instead of up and down it. Each row acts as a small barrier that slows running rainwater, so the water soaks into the soil instead of washing soil away. This reduces erosion, keeps moisture in the field and protects the topsoil.

Short Questions

3. What is the importance of rainwater harvesting in agriculture?

Rainwater harvesting collects rain from rooftops, fields or catchments and stores it in tanks, ponds or the soil for later use. It is important because it provides irrigation water in dry periods, recharges groundwater, reduces runoff and flooding, and lets rain-fed farmers grow a crop even when the rains fail. In water-scarce Pakistan it is one of the cheapest ways to secure water.

Short Questions

4. How does mulching contribute to soil and water conservation?

Mulching covers the soil surface with straw, leaves, crop residues or plastic. The cover shades the soil, so evaporation falls and moisture stays longer; it breaks the force of raindrops, so erosion is reduced; it suppresses weeds; and as organic mulch decays it adds humus, improving soil structure and fertility. The book calls mulching the soil's blanket for exactly these reasons.

Long Questions

1. Explain the modern concept of soil and water conservation. How does it differ

from traditional methods?

The modern concept treats soil and water as one connected system to be managed for long-term sustainability, not just for this year's crop. It uses scientific tools: soil testing, contour and precision farming, no-till and minimum tillage, drip irrigation, rainwater harvesting, watershed management and afforestation, all planned from data. Traditional methods relied on experience alone, flood irrigation, heavy ploughing and fallowing; they often wasted water, exposed the soil and degraded it over time. Modern conservation therefore aims at the same goal as traditional care of the land, but with measured, efficient and sustainable techniques.

Long Questions

2. Discuss the importance of watershed management. What strategies can be used to improve watershed health?

A watershed is the area of land that drains into one river or stream, and managing it well protects the water, soil and livelihoods of everyone downstream. Its importance is that it controls floods, recharges groundwater, prevents siltation of dams and canals, and keeps water clean and available. Strategies to improve watershed health are: afforestation and grass planting on slopes to hold soil; building check dams and ponds to slow runoff; contour farming and terracing; protecting stream banks with vegetation; and involving local communities in planning and maintenance, so the whole catchment is cared for together.

Long Questions

3. Describe the role of afforestation in soil and water conservation. How does it benefit watershed management?

Afforestation, planting trees on bare or degraded land, protects soil because tree

roots bind the soil and leaf litter covers it, cutting erosion by wind and water. Trees slow rainfall, increase infiltration and recharge groundwater, and their shade reduces evaporation. For watershed management this means steadier stream flows, less silt washing into rivers and reservoirs, cleaner water and better recharge of wells. Shelterbelts of trees also protect crops from hot, dry winds. Afforestation is therefore one of the most effective long-term conservation measures.

Inquisitive Questions

1. If you were to design a rainwater harvesting system for a rural area, what components would you include and why?

Model answer: my system would include a catchment area, such as rooftops or a cleaned, sloping field surface, to collect the rain; gutters and pipes to carry it without loss; a first-flush diverter and a simple filter to remove dust, leaves and debris; a storage tank or a lined pond sized to the village's dry-season need; and a distribution system of pipes or channels to the fields. I would add a recharge pit so surplus water soaks into the groundwater, and a community committee to maintain the system. Each part matters: without a clean catchment the water is dirty, without storage the rain is lost, and without maintenance the system soon fails.

Inquisitive Questions

2. How can modern technology be used to improve the efficiency of irrigation systems in agriculture?

Model answer: modern technology improves irrigation through soil-moisture sensors that switch water on only when the soil needs it, weather forecasts that stop irrigation before rain, and drip and sprinkler systems that deliver water directly to the roots with little loss. Solar pumps cut energy costs, laser

land-levelling makes flood irrigation even, and mobile apps let farmers control pumps remotely and schedule water by crop stage. Together these tools can save large amounts of water while giving crops exactly what they need, raising both yields and water productivity.

Chapter 6: Management of Soil and Water Problems

(A) Multiple Choice Questions, Tick the correct answer

1. The primary purpose of leaching in soil management is to _____

(a) Add nutrients to the soil

(b) Wash away excess salts from the soil ✓

(c) Increase soil compaction

(d) Reduce soil erosion

Correct answer: Wash away excess salts from the soil

(A) Multiple Choice Questions, Tick the correct answer

2. Which crop is suitable for growing in saline soils?

(a) Rice

(b) Barley ✓

(c) Sugarcane

(d) Cotton

Correct answer: Barley

(A) Multiple Choice Questions, Tick the correct answer

3. What is the main cause of waterlogging in soils?

(a) Excessive use of fertilizers

(b) Poor drainage systems ✓

(c) Overuse of pesticides

(d) High soil pH

Correct answer: Poor drainage systems

(A) Multiple Choice Questions, Tick the correct answer

4. Which of the following is a mechanical method to control soil erosion?

(a) Mulching

(b) Terracing ✓

(c) Crop rotation

(d) Agroforestry

Correct answer: Terracing

(A) Multiple Choice Questions, Tick the correct answer

5. What is the role of gypsum in reclaiming sodic soils?

(a) It adds nitrogen to the soil

(b) It replaces sodium with calcium ✓

(c) It increases soil acidity

(d) It reduces soil compaction

Correct answer: It replaces sodium with calcium

(A) Multiple Choice Questions, Tick the correct answer

6. Which practice helps reduce water loss through evaporation?

(a) Deep plowing

(b) Mulching ✓

(c) Overgrazing

(d) Frequent tilling

Correct answer: Mulching

(A) Multiple Choice Questions, Tick the correct answer

7. What is the primary benefit of using organic manure in soil reclamation?

(a) It increases soil salinity

(b) It improves soil structure and water retention ✓

(c) It reduces soil pH

(d) It causes soil compaction

Correct answer: It improves soil structure and water retention

(A) Multiple Choice Questions, Tick the correct answer

8. Which of the following is a cause of soil compaction?

(a) Adding organic matter

(b) Using drip irrigation

(c) Heavy machinery

(d) Planting cover crops

Correct answer: Heavy machinery

(B) Write Short Answers

1. What is the role of gypsum in reclaiming saline soils?

Gypsum (calcium sulphate) reclaims sodic and saline-sodic soils by supplying calcium, which replaces sodium on the soil particles. The displaced sodium forms soluble sodium sulphate, which is then washed down and out of the root zone by leaching with good-quality water. The soil structure opens up again, water moves through it, and crops can grow. Gypsum is therefore the standard chemical amendment for sodic soils.

(B) Write Short Answers

2. How does mulching help in water conservation?

Mulching covers the soil with straw, leaves or plastic, shading it from the sun and wind, so evaporation from the surface falls sharply and soil moisture lasts much longer between irrigations. It also slows runoff, letting rain soak in, and suppresses weeds that would otherwise drink the soil water. The result is more water kept in the root zone for the crop.

(B) Write Short Answers

3. What are the causes of soil compaction?

Soil compaction is mainly caused by the repeated passage of heavy machinery,

tractors and harvesters, especially on wet soil; by overgrazing, as animals' hooves press the soil; by continuous tillage at the same depth, which forms a hard pan; and by loss of organic matter, which makes the soil less springy. Compacted soil has few pores, so water and air cannot enter and roots cannot spread.

(B) Write Short Answers

4. What is the significance of contour plowing in soil conservation?

Contour ploughing means ploughing and sowing along the contours, the level lines across a slope, instead of up and down it. Its significance is that each furrow acts as a small dam: it slows rainwater running down the slope, lets it soak into the soil, and traps soil particles that would otherwise be washed away. Erosion falls greatly and the field keeps both its soil and its moisture.

(B) Write Short Answers

5. How does organic manure improve soil fertility?

Organic manure, such as farmyard manure and compost, improves fertility in several ways: it adds nitrogen, phosphorus, potassium and micronutrients directly; it increases humus, which holds nutrients and water in the soil; it feeds soil microbes that release nutrients slowly to plants; and it improves soil structure, making clay soils looser and sandy soils better at holding water. Its effect lasts longer than a dose of chemical fertiliser alone.

(B) Write Short Answers

6. What is the purpose of installing drainage systems in waterlogged soils?

Drainage systems, surface drains or underground tile drains, remove excess water from waterlogged soils so that air can return to the root zone. Without drainage, roots suffocate, salts accumulate as the water evaporates, and crops

fail. Good drainage lowers the water table, washes salts down below the roots, and restores the soil to productive cropping.

(C) Long Questions

1. Explain the methods used to reclaim saline and sodic soils.

Saline soils, which contain excess soluble salts, are reclaimed mainly by leaching: applying good-quality water in excess so the salts dissolve and wash below the root zone, with drainage provided to carry the salty water away. Sodic soils, which contain excess exchangeable sodium, need a chemical amendment first: gypsum is applied to replace sodium with calcium, and then leaching washes the sodium out. In both cases organic manure is added to restore structure, salt-tolerant crops such as barley are grown at first, and the field is levelled so water spreads evenly. The chapter also lists amendments such as sulphur for alkaline soils, superphosphate for phosphorus, urea for nitrogen and muriate of potash for potassium, chosen according to the soil's deficiency.

(C) Long Questions

2. Discuss the causes of soil erosion, and describe agronomic and mechanical practices to control it.

Soil erosion is caused by water running over bare slopes, wind blowing over dry, unprotected soil, overgrazing, deforestation, and careless tillage up and down slopes. Agronomic practices to control it are: cover crops and mulching to protect the surface, crop rotation and strip cropping, contour farming, and agroforestry shelterbelts that break the wind. Mechanical practices are: terracing steep slopes into level steps, contour bunding and check dams to slow water, and gully control structures. Agronomic methods protect the soil surface cheaply, while mechanical methods reshape the land where slopes are steep.

(C) Long Questions

3. How can farmers manage water loss in agriculture? Explain the role of drip irrigation, rainwater harvesting, and mulching in water conservation.

Farmers can manage water loss by reducing evaporation, runoff and deep percolation, and by applying water only where and when the crop needs it. Drip irrigation delivers water drop by drop directly to the root zone, cutting evaporation and runoff and saving up to half the water of flood irrigation. Rainwater harvesting stores rain in tanks or ponds for dry periods and recharges groundwater, so no rain is wasted. Mulching covers the soil and shades it, greatly reducing evaporation and keeping moisture in the root zone. Used together, and with laser levelling and scheduling by soil moisture, these three practices let a farmer grow more crop per drop of water.

(C) Long Questions

4. Describe the role of organic and chemical amendments in improving soil health. Provide examples of each and explain their benefits and limitations.

Organic amendments are materials of plant or animal origin, such as farmyard manure, compost and green manure. They add nutrients slowly, build humus, improve structure and water retention, and feed soil life; their limitations are that they are bulky, act slowly, and may not supply enough nutrients for high yields on their own. Chemical amendments are manufactured or mined materials used to correct a specific problem: gypsum to replace sodium in sodic soils, sulphur to lower the pH of alkaline soils, superphosphate to add phosphorus, urea to add nitrogen, and muriate of potash to add potassium. They act quickly and precisely, but overuse can cause salinity, acidity or pollution, and they do not build humus. The best practice is to combine both: chemical amendments correct the immediate deficiency, while organic amendments rebuild lasting soil health.

Chapter 7: Farm Forestry / Agro Forestry

Multiple Choice Questions (MCQs)

1. What is the main purpose of agroforestry?

(a) To grow only trees for timber

(b) To combine trees, crops, and livestock for sustainable farming ✓

(c) To replace all crops with trees

(d) To focus only on livestock farming

Correct answer: To combine trees, crops, and livestock for sustainable farming

Multiple Choice Questions (MCQs)

2. Which of the following is a benefit of agroforestry?

(a) Increased soil erosion

(b) Reduced biodiversity

(c) Improved soil fertility ✓

(d) Higher water evaporation

Correct answer: Improved soil fertility

Multiple Choice Questions (MCQs)

3. What is alley cropping?

(a) Growing crops in forested areas

(b) Planting trees in rows alongside crops ✓

(c) Raising livestock under tree canopies

(d) Planting trees around farm boundaries

Correct answer: Planting trees in rows alongside crops

Multiple Choice Questions (MCQs)

4. Which tree is known for nitrogen fixation in agroforestry?

(a) Mango

(b) Acacia ✓

(c) Eucalyptus

(d) Neem

Correct answer: Acacia

Multiple Choice Questions (MCQs)

5. What is the purpose of windbreaks in agroforestry?

(a) To increase soil erosion

(b) To protect crops from strong winds ✓

(c) To reduce water conservation

(d) To provide fodder for livestock

Correct answer: To protect crops from strong winds

Multiple Choice Questions (MCQs)

6. What is a key advantage of agroforestry for farmers?

(a) Reduced income sources

(b) Increased reliance on a single crop

(c) Diversified income from tree products ✓

(d) Higher risk of crop failure

Correct answer: Diversified income from tree products

Multiple Choice Questions (MCQs)

7. What is the Taungya system?

(a) Growing crops between young trees until the canopy closes ✓

(b) Planting trees around farm boundaries

(c) Combining livestock with forest farming

(d) Growing medicinal plants in home gardens

Correct answer: Growing crops between young trees until the canopy closes

Short Questions

1. What is the difference between farm forestry and agroforestry?

Farm forestry means growing trees on the farm, usually in blocks or woodlots, mainly for timber, fuelwood and other tree products. Agroforestry is wider: it deliberately combines trees with crops and/or livestock on the same land, so trees, crops and animals interact and benefit each other. Farm forestry produces trees; agroforestry produces trees plus crops plus livestock from one integrated system.

Short Questions

2. Name two benefits of trees in agroforestry systems.

Two benefits are: (1) trees improve the soil, because their leaf litter adds organic matter and nitrogen-fixing trees add nitrogen, while their roots bring up nutrients from deep layers; (2) trees protect crops and livestock, because windbreaks reduce wind damage and evaporation, and shade shelters animals from heat. Other benefits are extra income from fruit, timber and fodder, and habitat for wildlife.

Short Questions

3. What is the purpose of riparian buffers in agroforestry?

Riparian buffers are strips of trees, shrubs and grass planted along rivers, streams and canals. Their purpose is to trap sediment and farm chemicals before they reach the water, stabilise the banks against erosion, shade the water, and provide habitat for wildlife. They protect water quality while also giving the farmer fuelwood, fruit and fodder from the buffer trees.

Short Questions

4. How does agroforestry help combat climate change?

Agroforestry combats climate change mainly by storing carbon: trees absorb carbon dioxide from the air and lock it in wood, roots and soil organic matter for decades. It also reduces emissions, because less chemical fertiliser is needed where nitrogen-fixing trees enrich the soil, and it makes farms more resilient to climate shocks, as shade and windbreaks protect crops from heat, drought and storms.

Short Questions

5. What is the role of nitrogen-fixing trees in agroforestry?

Nitrogen-fixing trees, such as acacia and other leguminous trees, carry bacteria in root nodules that convert atmospheric nitrogen into plant-usable forms. Their fallen leaves and roots enrich the soil with nitrogen, so neighbouring crops need less nitrogen fertiliser, grow better and cost less to produce. They are the living fertiliser factories of an agroforestry system.

Short Questions

6. Why is tree species selection important in agroforestry?

Species selection decides whether the system succeeds or fails. The right tree must suit the local climate and soil, grow at a rate that does not shade out the crop, have roots that do not compete too strongly for water, and give products the farmer can sell, such as fruit, fodder or timber. A wrong choice, such as a thirsty, fast-spreading tree beside a crop, can reduce crop yields instead of raising them.

Long Questions

1. Explain the concept of agroforestry and its importance in sustainable farming. Provide examples of agroforestry practices.

Agroforestry is a sustainable land-use system that deliberately combines trees

with crops and/or livestock on the same land, so that the components interact ecologically and economically. It is important for sustainable farming because it raises total production from the same land, diversifies the farmer's income, restores soil fertility, conserves water, stores carbon and supports biodiversity, all at once. Examples: alley cropping, crops grown between rows of trees; silvopasture, livestock grazing under tree cover; windbreaks and shelterbelts protecting fields; home gardens mixing fruit trees, vegetables and poultry; and the Taungya system, where crops are grown between young forest trees until their canopy closes.

Long Questions

2. Discuss the advantages of alley cropping in agroforestry.

Alley cropping grows crops in the alleys between rows of trees, usually nitrogen-fixing or fruit trees. Its advantages are: the trees add nitrogen and leaf litter, improving soil fertility for the alley crops; tree roots reduce erosion and bring up deep nutrients; the rows act as windbreaks, cutting wind damage and evaporation; the farmer earns from both the alley crop and the tree products, fruit, fodder or timber; and the system supports more wildlife than a bare field. The main care needed is pruning the trees so they do not shade the crop too much.

Long Questions

3. Describe the role of agroforestry in improving soil fertility and water conservation. Use examples to support your answer.

Agroforestry improves soil fertility because tree leaf litter decays into humus, nitrogen-fixing trees such as acacia add nitrogen, and deep tree roots recycle nutrients from lower soil layers to the surface. For example, alley cropping with leguminous trees can cut the fertiliser a maize crop needs. For water

conservation, tree canopies break rainfall and reduce evaporation, roots open channels that let water soak in, and windbreaks slow drying winds, so soil moisture lasts longer; contour hedgerows of trees on slopes also trap runoff and soil. Together these effects keep the soil fertile and moist with fewer external inputs.

Long Questions

4. How can agroforestry contribute to the economy of Pakistan? Discuss its potential benefits and challenges.

Agroforestry can contribute to Pakistan's economy by raising farm incomes through diversified products, fruit, timber, fuelwood, fodder and medicinal plants, alongside crops; by supplying raw material to the timber and fruit industries and reducing timber imports; and by creating rural jobs in nurseries, planting and processing. It also protects the farmland itself from erosion and salinity, saving future production. Its challenges are: farmers need training and quality saplings, trees take years to pay back, insecure land tenure discourages planting, and trees can compete with crops if species and spacing are wrong. With extension support, secure tenure and markets for tree products, these challenges can be overcome.

Inquisitive Questions

1. If you were to design an agroforestry system for a dry region, what trees and crops would you choose and why?

Model answer: for a dry region I would choose drought-tolerant, deep-rooted trees such as acacia (kikar) and ber, which fix nitrogen, give fodder and fuelwood, and survive on little water, planted as windbreaks and in wide-spaced alleys. Between them I would grow drought-tolerant crops: barley, pulses such as gram, and oilseeds, with mulching to hold moisture. I would avoid thirsty trees

like eucalyptus near the crop, add a rainwater harvesting pit, and keep livestock, goats, to use the tree fodder. This combination survives drought, protects the soil and still gives food, fodder and income.

Inquisitive Questions

2. How can agroforestry help address food security and environmental challenges in your community?

Model answer: agroforestry addresses food security by producing food from several sources on the same land, fruit and nuts from trees, grain and vegetables from alleys, and milk and meat from livestock under the trees, so a failure of one component does not mean hunger. It addresses environmental challenges by stopping soil erosion, storing carbon, conserving water and providing habitat for pollinators and wildlife; the book notes that agroforestry systems can host up to 50 percent more wildlife species than conventional farms. In a community it also gives a living savings account, the tree bank idea: fast-growing fruit trees for short-term income and slow timber trees as a long-term investment.

Chapter 8: Production of Farm Animals

Multiple Choice Questions (MCQs)

1. What is the primary purpose of milch animals?

(a) Meat production

(b) Milk production ✓

(c) Draught power

(d) Wool production

Correct answer: Milk production

Multiple Choice Questions (MCQs)

2. Which of the following is an example of a draught animal?

(a) Cow

(b) Goat

(c) Ox ✓

(d) Chicken

Correct answer: Ox

Multiple Choice Questions (MCQs)

3. What is the main advantage of the intensive livestock production system?

(a) Low input costs

(b) High productivity ✓

(c) Use of natural grazing

(d) Low labor requirements

Correct answer: High productivity

Multiple Choice Questions (MCQs)

4. Which disease is common in poultry?

(a) Foot-and-mouth disease

(b) Avian influenza ✓

(c) Brucellosis

(d) Mastitis

Correct answer: Avian influenza

Note on Question 4

Note: why are the other options not poultry diseases?

Foot-and-mouth disease attacks cloven-hoofed animals such as cattle and buffaloes, brucellosis attacks cattle, buffaloes and goats, and mastitis is an udder infection of dairy animals. Avian influenza, bird flu, is the disease of poultry among the options, so option (b) is correct.

Multiple Choice Questions (MCQs)

5. What is a major challenge in animal breeding?

(a) Lack of market access

(b) Inadequate record-keeping ✓

(c) Overproduction of fodder

(d) Excessive use of antibiotics

Correct answer: Inadequate record-keeping

Multiple Choice Questions (MCQs)

6. Which livestock production system involves seasonal migration of animals?

(a) Intensive system

(b) Nomadic system ✓

(c) Zero-grazing system

(d) Commercial system

Correct answer: Nomadic system

Multiple Choice Questions (MCQs)

7. What is the primary cause of fodder shortage?

(a) Overproduction of crops

(b) Climate change and droughts ✓

(c) Excessive use of fertilizers

(d) Lack of veterinary services

Correct answer: Climate change and droughts

Multiple Choice Questions (MCQs)

8. What is the economic importance of livestock in Pakistan?

(a) It contributes 3% to GDP

(b) It provides employment to over 8 million families ✓

(c) It is the largest exporter of dairy products

(d) It has no significant impact on the economy

Correct answer: It provides employment to over 8 million families

Multiple Choice Questions (MCQs)

9. Which of the following is a problem in animal marketing?

(a) High demand for livestock products

(b) Dominance of middlemen ✓

(c) Excessive use of modern technology

(d) Overproduction of fodder

Correct answer: Dominance of middlemen

Short Questions

1. What are the two main purposes of raising farm animals?

The two main purposes are: (1) production of food and products, milk from milch animals, meat from beef animals and poultry, eggs from layers, and wool and

hides; and (2) work and other services, draught power for ploughing and transport from oxen and other draught animals, and manure for fertilising fields. Farm animals thus serve both as a source of food and income and as a source of power and farm inputs.

Short Questions

2. Name two common diseases in cattle and buffaloes.

Two common diseases are foot-and-mouth disease, which causes blisters on the mouth and feet, fever and a sharp fall in milk, and mastitis, an infection of the udder that reduces milk yield and quality. Other common diseases are brucellosis, haemorrhagic septicaemia and black quarter; all are controlled by vaccination, hygiene and prompt veterinary treatment.

Short Questions

3. What is the difference between the extensive and intensive livestock production systems?

In the extensive system animals graze over large areas of natural pasture with low inputs, low labour and low output per animal, as in nomadic and free-range systems. In the intensive system animals are kept in sheds, fed formulated rations and managed closely with high inputs of feed, housing, health care and labour, giving high productivity per animal, as in commercial dairy and poultry farms. Extensive farming is cheap but low-yielding; intensive farming is costly but high-yielding.

Short Questions

4. How does climate change affect fodder availability?

Climate change brings higher temperatures, erratic rainfall and more frequent droughts, which reduce the growth of fodder crops and pastures and shorten the

grazing season. Floods can wash fodder away, and heat stress also cuts the quality of fodder. The result is fodder shortage, higher feed prices and weaker, less productive animals, especially in arid and rain-fed areas.

Short Questions

5. Why is animal health important for sustainable farming?

Healthy animals give more milk, meat and eggs, breed regularly and live longer, so the farmer earns more from the same feed and housing. Sick animals waste feed, need costly treatment, spread disease to the herd, and may pass diseases to people. Good health, through vaccination, hygiene, balanced feeding and clean housing, is therefore essential for profitable and sustainable farming.

Long Questions

1. Explain the economic importance of livestock in Pakistan. Discuss its contribution to food security and rural livelihoods.

Livestock is a pillar of Pakistan's rural economy: it provides milk, meat and eggs that form a large part of the nation's food, and it gives employment to over 8 million families, as the chapter's own MCQ states. For food security, livestock supplies daily protein, especially milk, to families that may own little or no land, and animals can be sold in an emergency as a living bank. For rural livelihoods, livestock gives a regular cash income from daily milk sales, unlike seasonal crops, provides draught power and manure, and supports related trades in feed, veterinary care, transport and processing. It is therefore central to rural incomes and nutrition.

Long Questions

2. Describe the challenges faced in animal breeding and suggest solutions to improve breeding practices.

Breeding challenges are: inadequate record-keeping, so good animals cannot be identified; use of low-quality, unproven bulls and inbreeding in small herds; poor nutrition and health, which hide an animal's true genetic merit; and weak artificial-insemination services in remote areas. Solutions are: keeping simple records of milk yield, calving and parentage for every animal; using proven, pedigree bulls and artificial insemination with tested semen; selecting breeding stock for milk, growth and disease resistance; avoiding mating of close relatives; and supporting farmers with veterinary extension, feed advice and breed-improvement programmes.

Long Questions

3. Discuss the role of the nomadic farming system in Pakistan. What are its major problems, and how can they be addressed?

In the nomadic system, herders move their flocks of sheep, goats and camels seasonally between summer highland pastures and winter lowland areas, using land that cannot support settled farming. Its role is to turn sparse rangeland vegetation into meat, milk, wool and income, and to sustain pastoral communities and their culture. Its major problems are shrinking grazing lands, water shortage on migration routes, lack of veterinary services, schools and markets for mobile families, and conflict with settled farmers. They can be addressed by protecting grazing routes and water points, providing mobile veterinary and extension services, giving herders market access and fair prices, and insuring flocks against drought and disease.

Long Questions

4. Explain the importance of conserving good-quality animal breeds through genetic means. How can this be achieved?

Good local breeds, such as Sahiwal cattle and Nili-Ravi buffaloes, are adapted to

Pakistan's heat, diseases and poor fodder, and they carry valuable genes for milk, hardiness and fertility. If they are crossed indiscriminately or neglected, these genes are lost forever and future breeding has nothing to build on. Conservation is achieved by identifying and registering pure animals, keeping pedigree and performance records, maintaining nucleus herds and semen banks of superior bulls, using selective breeding within the breed rather than random crossing, and paying farmers to keep purebred animals.

Inquisitive Questions

1. If you were to design a livestock production system for a rural area, what type of system would you choose and why?

Model answer: for a typical rural area I would choose a semi-intensive system: animals housed at night and grazed or stall-fed during the day on grown fodder, with a small commercial dairy or poultry unit as the cash enterprise. It combines the low cost of grazing with the high productivity of controlled feeding and health care, suits small landholdings, and gives daily milk income plus manure for the fields. Pure extensive grazing gives too little income, and a fully intensive system needs capital and feed that most small farmers cannot afford.

Inquisitive Questions

2. How can modern technology be used to solve the problems of fodder shortage and animal health in livestock production?

Model answer: fodder shortage can be eased by high-yielding, drought-tolerant fodder varieties, hydroponic fodder units that grow green feed in days, silage making to preserve surplus fodder, and mobile advisories that tell farmers when to sow fodder. Animal health can be improved by vaccination schedules sent by mobile phone, telemedicine with veterinarians, diagnostic kits for early disease detection, and record-keeping apps that track each animal's health, breeding

and milk. Together these tools cut losses from hunger and disease and raise productivity.

Inquisitive Questions

3. Do you think the nomadic farming system can coexist with modern agricultural practices? Why or why not?

Model answer: yes, it can coexist if both sides adapt. Modern practices can serve nomads through mobile veterinary services, solar-powered cold storage for milk on migration routes, mobile phone market prices, and improved, drought-tolerant fodder planted along routes. In return, nomadic grazing can fit modern sustainable farming, because it uses rangelands that cannot be cropped and recycles nutrients naturally. Coexistence fails only where grazing land is fenced off without alternatives or where nomads get no services; with protected routes, services and fair markets, the two systems complement each other.

Chapter 9: Dairy Industry

Multiple Choice Questions (MCQs)

1. What is the main purpose of the dairy industry in Pakistan?

(a) Producing crops

(b) Raising birds for meat

(c) Producing milk and dairy products ✓

(d) Making leather

Correct answer: Producing milk and dairy products

Multiple Choice Questions (MCQs)

2. Which province in Pakistan is the largest producer of milk?

(a) Sindh

(b) Punjab ✓

(c) Khyber Pakhtunkhwa

(d) Balochistan

Correct answer: Punjab

Multiple Choice Questions (MCQs)

3. What is the main cause of milk spoilage during storage?

(a) Lack of sunlight

(b) High temperature ✓

(c) Overfeeding animals

(d) Using glass bottles

Correct answer: High temperature

Multiple Choice Questions (MCQs)

4. Which disease in dairy animals can reduce milk production?

(a) Foot-and-mouth disease ✓

(b) Newcastle disease

(c) Coccidiosis

(d) Aspergillosis

Correct answer: Foot-and-mouth disease

Note on Question 4

Note: why are the other options not dairy diseases?

Newcastle disease attacks poultry, coccidiosis mainly attacks poultry and young animals' intestines, and aspergillosis is a fungal disease of birds and stored feed. Foot-and-mouth disease attacks cattle and buffaloes, causes fever and blisters, and sharply reduces milk production, so option (a) is correct.

Multiple Choice Questions (MCQs)

5. What is the purpose of pasteurization in milk processing?

(a) To add flavor to milk

(b) To kill harmful bacteria ✓

(c) To increase fat content

(d) To change milk color

Correct answer: To kill harmful bacteria

Multiple Choice Questions (MCQs)

6. Which equipment is used to separate cream from milk?

(a) Milking machine

(b) Cream separator ✓

(c) Homogenizer

(d) Pasteurizer

Correct answer: Cream separator

Multiple Choice Questions (MCQs)

7. What is the main benefit of using modern technology in dairy farming?

(a) Increases manual labor

(b) Reduces milk production

(c) Improves efficiency and quality ✓

(d) Makes animals grow slower

Correct answer: Improves efficiency and quality

Multiple Choice Questions (MCQs)

8. Which dairy product is made by fermenting milk?

(a) Butter

(b) Cheese

(c) Yogurt ✓

(d) Ghee

Correct answer: Yogurt

Multiple Choice Questions (MCQs)

9. What is the role of cold chain systems in the dairy industry?

(a) To increase milk spoilage

(b) To keep milk fresh during transport ✓

(c) To reduce milk production

(d) To add water to milk

Correct answer: To keep milk fresh during transport

Multiple Choice Questions (MCQs)

10. Which of the following is a value-added dairy product?

(a) Raw milk

(b) Flavored milk ✓

(c) Unpasteurized milk

(d) Spoiled milk

Correct answer: Flavored milk

Short Questions

1. What are the two main types of dairy industries in Pakistan?

The two main types are the traditional, informal sector and the modern, formal sector. The informal sector consists of small farmers and milkmen (doodhis) who sell raw, loose milk directly to consumers, and it handles most of Pakistan's milk. The formal sector consists of organised dairy companies that collect, pasteurise, package and process milk into UHT milk, yogurt, cheese, butter and other products.

Short Questions

2. Name two common diseases that affect dairy animals.

Two common diseases are foot-and-mouth disease, which causes fever and blisters and a sharp drop in milk, and mastitis, an infection of the udder that reduces milk yield and spoils milk quality. Both are controlled by vaccination, clean housing and milking hygiene, and prompt veterinary treatment.

Short Questions

3. What is the purpose of a cream separator in milk processing?

A cream separator spins milk at high speed in a centrifuge, so the lighter cream separates from the heavier skimmed milk. Its purpose is to obtain cream for making butter, ghee and other products, while the skimmed milk is used for drinking, yogurt or milk powder. It works faster and more completely than letting cream rise naturally.

Short Questions

4. How does modern technology help in dairy farming?

Modern technology helps through milking machines that milk quickly and hygienically, milk chillers and cold chains that keep milk fresh, automated feeding and health-monitoring systems, artificial insemination with superior semen, and testing kits that check milk quality. Together these raise milk yield and quality, cut labour and spoilage, and give farmers better prices.

Short Questions

5. What are the benefits of value addition in dairy products?

Value addition, turning raw milk into yogurt, cheese, butter, flavoured milk, ice cream, khoya sweets, condensed and powdered milk, and whey products, increases the farmer's and processor's income, because processed products sell for more than raw milk. It extends shelf life, reduces spoilage losses, creates jobs in processing and packaging, and gives consumers a wider, safer choice of products.

Long Questions

1. Explain the challenges faced by the dairy industry in Pakistan and suggest measures to overcome them.

The challenges are: most milk is sold raw and unchilled, so spoilage is high, especially in summer; small farmers lack chilling, transport and cold chains; middlemen take a large share, so farmers get low prices; animal diseases and poor fodder keep yields low; and adulteration and weak quality testing harm consumers. Measures to overcome them are: village milk collection and chilling centres linked by refrigerated transport; farmer cooperatives that bypass middlemen and pay fair prices; vaccination, better fodder and breeding programmes to raise yields; strict quality testing and action against adulteration;

and training and affordable credit for small dairy farmers.

Long Questions

2. Describe the role of modern technology in improving milk production and processing in the dairy sector.

Modern technology improves production through superior breeds and artificial insemination, balanced compound feed, automated milking machines, and health monitoring with sensors and records, all of which raise yield per animal. In processing it improves quality through pasteurisers that kill harmful bacteria, homogenisers that give smooth milk, cream separators, UHT plants that give long shelf life, and cold chains from farm to shop that prevent spoilage. Packaging innovations such as resealable caps, portion packs and eco-friendly materials, and quality certifications such as organic or antibiotic-free labels, further raise value and consumer trust.

Long Questions

3. Discuss the importance of proper packaging and value addition in dairy products. How do they benefit farmers and consumers?

Proper packaging protects milk from bacteria, light and contamination, keeps it fresh during transport, and carries labels showing the product, date and quality, which builds consumer trust. Value addition converts raw milk into yogurt, cheese, butter, ghee, flavoured milk, ice cream, khoya sweets and milk powder, which last longer and sell for more. Farmers benefit because processed products fetch higher, steadier prices and reduce losses from spoilage. Consumers benefit because they get safe, hygienic, convenient products in many varieties, available all year, instead of raw milk that must be used the same day.

Inquisitive Questions

1. Why is only 5% of Pakistan's milk processed and packaged, while the rest is sold raw? What can be done to increase the share of processed milk?

Model answer: only about 5 percent is processed because most milk comes from millions of small farmers with one or two animals, scattered in villages without chilling centres or refrigerated transport, so milk must be sold raw and quickly through local milkmen. Consumers also prefer fresh loose milk, which is cheaper, and processing plants are few and far from villages. The share can be increased by building village collection and chilling centres, organising farmer cooperatives, giving affordable credit for small chillers, running awareness campaigns on the safety of pasteurised milk, and keeping processed milk prices close to loose milk prices so consumers switch willingly.

Chapter 10: Poultry Husbandry

Multiple Choice Questions (MCQs)

1. What is the main purpose of poultry farming?

(a) Growing crops

(b) Raising birds for meat and eggs ✓

(c) Producing dairy products

(d) Making leather

Correct answer: Raising birds for meat and eggs

Multiple Choice Questions (MCQs)

2. Which province in Pakistan is the largest producer of poultry?

(a) Sindh

(b) Punjab ✓

(c) Khyber Pakhtunkhwa

(d) Balochistan

Correct answer: Punjab

Multiple Choice Questions (MCQs)

3. What is the purpose of vaccination in poultry?

(a) To increase egg size

(b) To protect birds from diseases ✓

(c) To make birds grow faster

(d) To change feather color

Correct answer: To protect birds from diseases

Multiple Choice Questions (MCQs)

4. Which feed is given to young chicks for their early growth?

(a) Layer feed

(b) Broiler feed

(c) Starter feed ✓

(d) Finisher feed

Correct answer: Starter feed

Multiple Choice Questions (MCQs)

5. Which modern technology is used to maintain temperature in poultry houses?

(a) Automated feeding systems

(b) Climate control systems ✓

(c) Robotic cleaners

(d) Genetic improvements

Correct answer: Climate control systems

Multiple Choice Questions (MCQs)

6. Which of the following is a fancy bird?

(a) Broiler chicken

(b) Silkie chicken ✓

(c) Layer hen

(d) Turkey

Correct answer: Silkie chicken

Short Questions

1. What is the difference between broiler and layer farming?

Broiler farming raises chickens for meat: birds are fed a high-energy ration, grow very fast, and are ready for market in about six weeks. Layer farming raises hens for eggs: birds are fed a calcium-rich layer ration, start laying at about five months, and lay eggs for a year or more. Broilers give a quick, one-time return; layers give a steady daily income from eggs.

Short Questions

2. Name two diseases that affect poultry birds.

Two common poultry diseases are Newcastle disease (Rani Khet), a viral disease that causes breathing difficulty, nervous signs and heavy deaths, and avian influenza (bird flu), which causes sudden deaths and is a danger to people as well. Other common diseases are coccidiosis, infectious bursal disease (Gumboro) and fowl pox; all are controlled by vaccination, hygiene and biosecurity.

Short Questions

3. Why is proper feed storage important in poultry farming?

Proper feed storage matters because damp, hot or dirty storage lets feed grow moulds and fungi, some of which produce toxins that poison birds, cut growth and egg laying, and kill chicks. Rodents and insects also eat and contaminate stored feed. Feed must therefore be stored dry, cool, ventilated and off the floor, in clean, covered bins, and used in rotation so old feed is used first.

Short Questions

4. What is the role of vaccination in preventing poultry diseases?

Vaccination exposes birds to a weakened or killed form of a disease germ, so their immune system learns to fight it without suffering the disease. When the real germ later arrives, vaccinated birds resist it, so outbreaks are prevented or mild, deaths fall, and growth and egg laying are protected. Vaccination on schedule, together with hygiene and biosecurity, is the cheapest and most effective disease control in poultry farming.

Long Questions

1. Explain the importance of poultry farming in Pakistan's economy.

Poultry farming is one of Pakistan's most important agricultural industries. It supplies cheap, high-quality protein, chicken meat and eggs, to a large population at prices poorer families can afford, strengthening food security. It provides employment to millions in breeding, hatcheries, feed mills, farming, processing, transport and sale, and it supports the maize and soybean industries that supply poultry feed. Because a poultry unit starts producing in weeks and needs little land, it is also an important income source for small farmers and women in rural and peri-urban areas.

Long Questions

2. Discuss the challenges faced by poultry farmers and suggest measures to overcome them.

Poultry farmers face disease outbreaks that can wipe out a flock, high and fluctuating feed prices, heat stress in summer, unreliable electricity for climate control, and market price crashes when many farmers sell at once. Measures to overcome them are: strict vaccination schedules and biosecurity, keeping new birds separate and houses clean; buying quality feed and storing it properly; climate-controlled housing with fans, cooling pads and backup power; contracts or cooperatives that stabilise prices; and training plus affordable credit and insurance for small farmers.

Long Questions

3. How can modern technology improve the production, processing, and marketing of poultry products?

In production, automated feeding and watering, climate control systems and sensors monitor temperature, humidity and bird health continuously, improving growth and egg laying while cutting labour. In processing, automated slaughtering, chilling, grading and packaging give hygienic meat and eggs with

longer shelf life. In marketing, the chapter lists modern techniques such as partnering with restaurants, supermarkets and hotels to reach a wider audience, quality certifications like organic or antibiotic-free labels that attract health-conscious buyers and justify higher prices, online platforms and social media to reach city buyers, and customer feedback and reviews that improve products and build trust. Together these raise sales, quality and competitiveness.

Inquisitive Questions

1. Why do some poultry farms use automated systems for feeding and watering birds? How does it help farmers and the birds?

Model answer: automated systems give every bird the right amount of feed and clean water at the right time, day and night, without the farmer having to be present. They help farmers by cutting labour, reducing feed waste and letting one worker manage thousands of birds. They help birds because feed and water are always fresh and available, so birds grow evenly, lay better and suffer less stress and disease than with irregular hand feeding.

Inquisitive Questions

2. What happens if birds are not vaccinated on time?

Model answer: if birds are not vaccinated on time, they have no immunity when a disease germ reaches the farm, so diseases such as Newcastle disease, Gumboro or bird flu can spread through the whole flock in days, causing heavy deaths, a collapse in egg laying, and large financial loss. Late vaccination may come after infection has already taken hold, when it can no longer save the birds. The whole flock, and neighbouring farms, are put at risk, and the farmer may lose his entire investment.